OWEN NYO WEI YUAN

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WORK EXPERIENCE

Software Engineer (Internship), Red Airship | Singapore **May 2026 - Present**

- Automated payment link creation, payment requests and email reminders in a client CRM, replacing a fully manual request-and-chase process.
- Maintained and extended the two-way data sync between an external booking platform and the client CRM, keeping booking records consistent across both systems.
- Led an AI pilot building a digital concierge with RAG, an MCP server and AWS Lambda, projected to cut online operator queries by 60%.
- Built frontend forms and a Playwright end-to-end test suite integrated into CI/CD, raising test coverage to 80%+ and cutting testing time by over 50%.

Teaching Assistant, SIT | Singapore **Sep 2025 - Apr 2026**

- Guided 150+ junior students in labs across 6 modules covering mathematics, programming and software design.
- Graded 150+ assignments per trimester and revised 14+ lab exercises with faculty to match course objectives.

Software Tester (Internship), Alexis Alliance | Singapore **Jan 2021 - Jun 2021**

- Designed and executed 20+ test cases and tracked 150+ defects with developers across the release cycle.

EDUCATION

Singapore Institute of Technology **Aug 2023 - Apr 2027 (expected)**

Bachelor of Engineering majoring in Software Engineering

- CGPA: 4.11 / 5.00
- Served as a mentor in the ICT-SMC mentoring programme for 3 consecutive years

Nanyang Polytechnic **Apr 2018 - May 2021**

Diploma in Infocomm and Security

- Awarded the Edusave Good Progress Award & Edusave Skills Award

CERTIFICATIONS

- Claude Certified Architect - Foundations — Jun 2026
- Professional Scrum Master I (PSM I) — Aug 2025
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional — Jul 2024
- Oracle Cloud Infrastructure 2023 AI Certified Foundations Associate — Jul 2024
- Oracle Cloud Infrastructure 2023 Certified Foundations Associate — Jun 2024

PUBLICATIONS

- Algorithmic Trading Strategy Development and Optimisation [arXiv preprint arXiv:2603.15848] — Mar 2026
- IEEE International Smart Cities Conference on “Smart Cities: Revolution for Mankind” (P06110) — Aug 2025

SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, Java, SQL, C#, C
- **AI Engineering:** RAG, MCP, Tool Calling, Agent Memory, Vector Databases, LLM Evaluation, NLP (FinBERT)
- **Architecture & Design:** API design (REST, GraphQL), Serverless architecture, Caching (Redis), Schema Design
- **Backend:** Node.js, NestJS, Express.js, FastAPI
- **Frontend:** React, Next.js, Redux, TypeScript
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), CI/CD (AWS CodePipeline, CodeBuild), Docker, Git, Upsun
- **Observability & Security:** AWS CloudWatch (logs, alarms), JWT, OWASP
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase, Firebase, Schema Design
- **Testing:** Playwright, E2E Testing, Test Automation
- **Spoken Languages:** English, Chinese